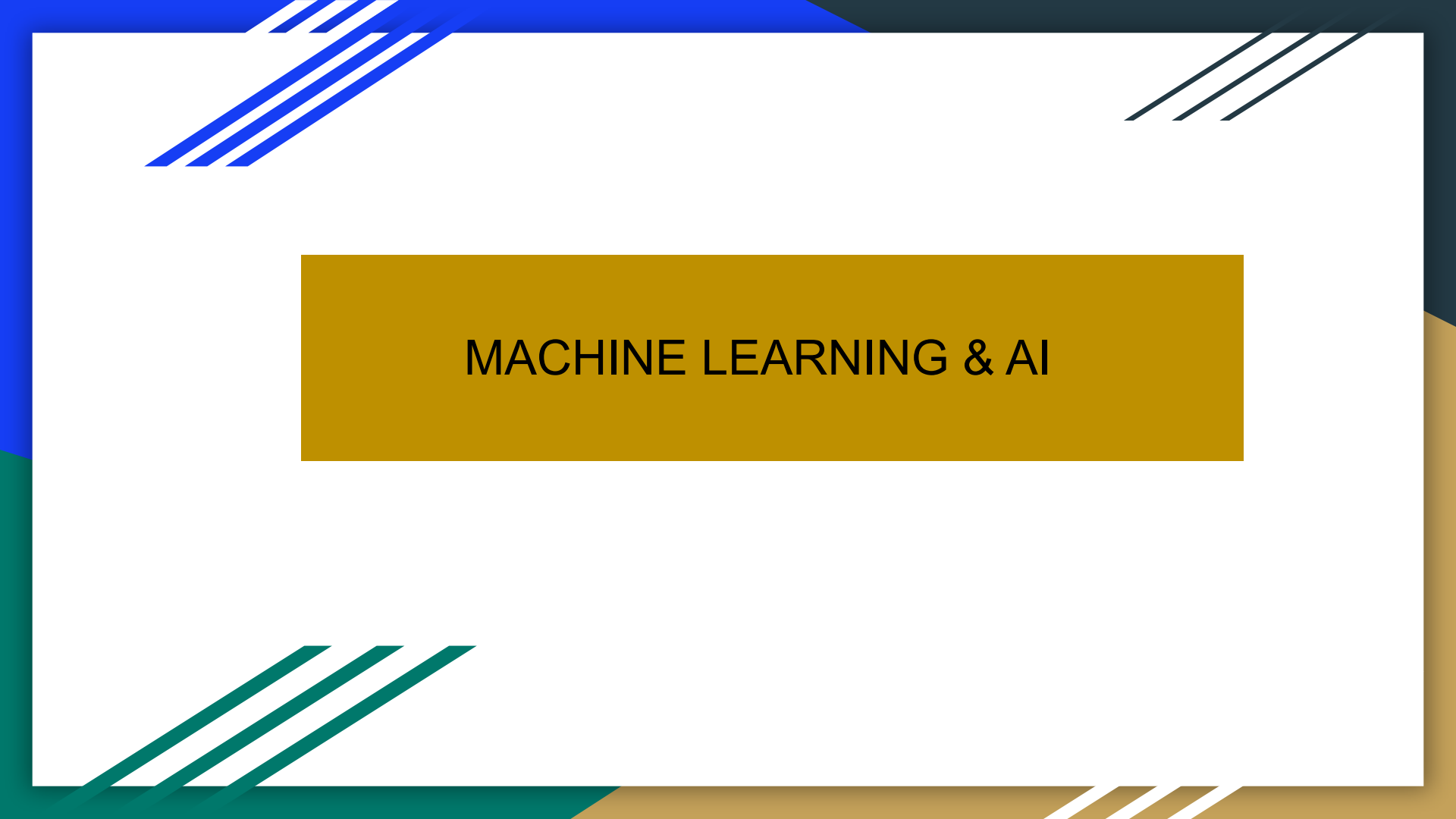




MACHINE LEARNING & AI

WEEK 4 – Major Project (Capstone Project)

Focus: Full ML pipeline + Documentation + Presentation

Choose **any one** of these major projects .

Option 1: Heart Disease Prediction

Dataset:

Kaggle – [Heart Disease UCI](#)

Workflow:

- Data cleaning + feature scaling
- Train Logistic Regression / Decision Tree
- Evaluate accuracy, precision, recall
- Confusion matrix & ROC curve visualization

Outcome: Predict if patient likely to have heart disease.

Option 2: Bank Loan Approval Prediction

Dataset:

[Kaggle – Loan Prediction Dataset](#)

Workflow:

- Clean categorical data (Gender, Marital Status, Income)
- Train Random Forest / XGBoost model
- Evaluate with Confusion Matrix
- Predict if loan can be approved

Outcome: Practical classification problem.

Option 3: Customer Segmentation using Clustering

Dataset:

[Kaggle – Mall Customers Dataset](#)

Workflow:

- Apply K-Means to cluster customers
- Visualize customer groups by Age/Income/Spending Score
- Label groups as “Low Spenders”, “Medium Spenders”, “High Spenders”

Outcome: Business analytics-oriented project.